# Neel Maddu

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## Education

**Georgia Institute of Technology** — Atlanta, GA

*Bachelor of Science in Computer Science* | ***GPA: 4.0*** — *Expected Dec 2027*

- **Coursework:** Machine Learning, Artificial Intelligence, Design & Analysis of Algorithms, Data Structures & Algorithms, Systems & Networks, Software Engineering, Objects & Design.
- **U.S. Citizen** | Eligible for security clearance.

## Experience

**Amazon** — Jan. 2027 – Apr. 2027

*Incoming Software Development Engineer Intern* — *Bellevue, WA*

- Joining Supply Chain Optimization Technologies (SCOT) **Fulfillment Optimization**, the team behind the Tier 1 services that decide how customer orders flow through Amazon's fulfillment network.

**Georgia Tech Research Institute (GTRI)** — May 2026 – Present

*Software Engineer Intern, CIPHER Lab – Threat Analysis Branch* — *Atlanta, GA*

- Build NLP pipelines for entity extraction and semantic analysis across **60K+** unstructured text records and posts, supporting cybersecurity and intelligence analysis workflows in a national security context.
- Evaluate **15 agentic configurations** against **6 categories** of adversarial and injection-style attacks across an isolated **3-VM** testbed, quantifying how prompt design, tool permissions, and guardrail placement drive failure rates.

**Georgia Institute of Technology – VIP Program** — Jan. 2026 – Present

*Undergraduate Researcher, Automated Algorithm Design* — *Atlanta, GA*

- Own the judge for an automated LLM jailbreaking pipeline built on LLM-Guided Evolution (LLM-GE), cited as prior work by Google DeepMind's AlphaEvolve; designed the scoring formula the team adopted as the search's fitness function.
- Proposed a two-judge setup with HarmBench as an independent reporting judge to catch the search overfitting its own scorer; first runs on the PACE ICE HPC cluster showed a **46.2% vs. 30.6%** attack-success gap between judges.

**Georgia Tech iOS Club** — Aug. 2025 – Present

*Tech Lead (Aug. 2026 – Present); previously Senior iOS Developer and iOS Developer* — *Atlanta, GA*

- Co-lead a **20+ developer** team building Contour, an on-device iOS accessibility app that guides blind and low-vision users to buttons on unfamiliar appliance panels using Vision and haptic/audio feedback.
- Restructured the shared Swift codebase for **3 sub-teams** and **12 parallel lanes** working against common interface contracts; maintain CI and tooling, including the migration to Xcode 27 and iOS 27.

**GTPoker (Georgia Tech Poker Club)** — Oct. 2026 – Present

*Director of Technology* — *Atlanta, GA*

- Lead GTAlgoPoker, the club's six-max Hold'em bot league on AWS: migrated it and the club site to the club's own account, rebuilt on S3, CloudFront, Lambda and an auto-pausing Aurora database in place of an always-on server.
- Hardened the bot runner: removed match tasks' AWS role, limited each bot to its own source via per-seat signed URLs, ran each bot as a separate user, and added an HMAC seat check, closing **13 of 32** code-review findings.

## Projects

**MARTA Live Transit Tracker** | *Python, FastAPI, SQLite, SwiftUI, WidgetKit, GTFS-Realtime* — 2026

- Built a full-stack transit tracker: a 24/7 FastAPI/SQLite backend polling MARTA's GTFS-Realtime and rail feeds, accumulating **575K+ arrival observations** across 78 routes and 6,200+ stops, feeding a native SwiftUI app tracking ∼200 live vehicles at 15-second refresh.
- Fixed undocumented feed defects (reconstructed missing bus delays via joins against **2.4M GTFS rows**; hand-wrote a protobuf decoder validated byte-for-byte against the reference parser) and integrated OpenTripPlanner for delay-aware trip planning; **59 automated tests**.

**Hidden Quakes (HackGT 13)** | *Python, PyTorch, SeisBench, PyOcto, ObsPy, React Three Fiber* — 2026

- Owned the seismology lane of a 4-person, 36-hour build that rebuilt a **15x denser** earthquake catalog for Utah's FORGE geothermal site from raw public waveforms: **654 candidate events vs. 43** public, with **43/43** catalog recall.
- Built PyOcto event association, grid-search relocation with station statics, quality tiering, and magnitude calibration (leave-one-out MAE 0.11) over 27K picks from 24 stations; a 20-shuffle null test found **0** strict-tier chance events.

**LLM Red-Teaming Evaluation Pipeline** | *Python, Streamlit, Ollama, Anthropic API, pytest* — 2026

- Built an extensible prompt-injection evaluation framework testing local and frontier LLMs against **46 attack payloads** across 41 techniques, with LLM-as-judge scoring; measured breakthrough rates of up to **47%** on local 7–8B models versus **3%** on a frontier model.

## Technical Skills

**Languages**: Python, Java, C, C++, JavaScript, Swift, SQL
**ML & AI**: PyTorch, TensorFlow, scikit-learn, FAISS, NLP, LLM Evaluation, Graph Analytics
**Frameworks & Libraries**: FastAPI, Node.js, React, Next.js, SwiftUI, Core ML, Vision, Streamlit, pandas, NumPy
**Data & Infrastructure**: MySQL, MongoDB, SQLite, Firebase, Docker, Git, GitHub Actions, Jenkins, CI/CD